Class: SE **Subject: COA LAB** **Sem-III**

Experiment No 8
Course Outcome: CO4 **Blooms Level: L4**
Aim: To develop Password Detection Application using DOS interrupts along with 8086 instructions.
Abstract: Develop assembly language program for 8086 to allow user to enter password for the application. If password is correct then display the message on standard output as "Password Matched" otherwise, display the message, "Password doesn't Matched". The program prompts the user to input a password through standard input, compares the entered value with a predefined password stored in memory, and displays the corresponding output message. If the password entered by the user matches the predefined value, the program outputs the message *"Password Matched"*. Otherwise, it displays *"Password doesn't Matched"*. This implementation demonstrates fundamental concepts of string handling, user interaction through DOS interrupts, and conditional branching in 8086 assembly programming.
Sample Input and Output: **Case 1:** **Input:** Enter your password: LIBRARY **Output:** *Password doesn't Matched* **Case 2:** **Input:** Enter your password: APSIT **Output:** *Password Matched*
Theory: An interrupt is a special condition that arises during working of a microprocessor. The microprocessor services it by executing a subroutine called Interrupt Service Routine (ISR). 1) Software interrupt: These interrupts are caused by writing the software interrupt instruction INT n where 'n' can be any value from 0 to 255 (00H to FFH). Hence, all 256 interrupts can be invoked by software. Software interrupt Interrupt types 0 - 1Fh are known as BIOS interrupts and interrupt types 20h-3Fh are serviced by DOS routines that provide high-level service to hardware as well as system resources. **INT 21H interrupt** This interrupt provides many functions for doing keyboard, video, Output display and file operations

i) INT 21H / AH=09H (Display a string on Monitor)

This interrupt is used to display string on output device like monitor. String to be displayed must be terminated by '$' character. DS must point to the string's segment, and DX must contain the string's offset. Suppose following string is stored in data segment.

MSG DB "Hello World 8086 $" ; msg to be displayed

To display a string on output device load DX with offset address of string, AH with 09H and call INT 21H interrupt.

```
LEA DX, MSG        ; Offset address of string
MOV AH, 09H        ; DOS function (Display String)
INT 21H
```

ii) INT 21H / AH=0AH (Accept the Input string from Keyboard)

Reading a string is accomplished by function 0AH, INT 21H. DOS function 0AH will accept a string of text entered at keyboard and copy that string into memory buffer. DOS 0AH is invoked with DS: DX, pointing to an input buffer, whose size should be at least three bytes longer than the largest input string anticipated.

Buffer Length	Number of characters Entered	----	----	----	----	----	0x0d

Keyboard Buffer structure

Buffer invoking DOS function 0AH, programmer must set the first byte of the buffer with the number of character size in the buffer. After returning from DOS function 0AH, the second byte of the buffer will contain a value giving the number of character actually read from keyboard.

To accept input string though keyboard, allocate memory as a buffer in data segment with specific offset address, load AH with 0AH and call INT 21H interrupt.

```
LEA DX, BUFFER     ; Offset address of Buffer
MOV AH, 0AH        ; DOS function
INT 21H
```

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An assembly Program to check password:

```
DISPLAY MACRO MSG

    LEA DX, MSG
   MOV AH, 09H
    INT 21H
ENDM
DATA SEGMENT
USERMSG  DB "ENTER YOUR PASSWORD: $"
CORRECT  DB  0AH, 0DH, "PASSWORD MATCHED$"
WRONG      DB  0AH, 0DH, "PASSWORD DOESN'T MATCHED$"
USER_INPUT DB 10
         DB 10 DUP (00H)
PASSWORD   DB "APSIT"
          DATA ENDS
CODE SEGMENT
ASSUME CS: CODE, DS: DATA
START:  MOV AX, DATA
          MOV DS, AX
          MOV ES, AX

          DISPLAY USERMSG

          LEA DX, USER_INPUT
          MOV AH, 0AH
          INT 21H

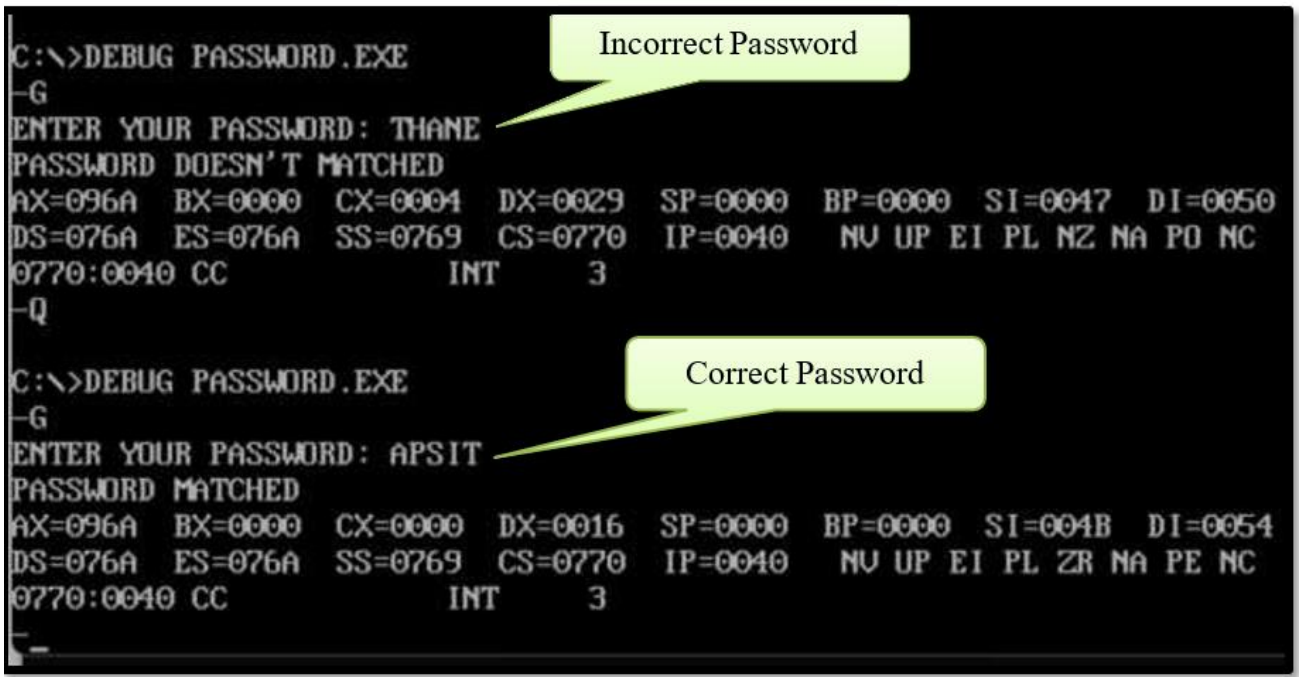
          MOV CL, USER_INPUT+1
          CMP CL, 05H
          JNZ UNMATCH

          LEA SI, USER_INPUT+2
           LEA DI, PASSWORD
          CLD
          REPZ CMPSB
          JNZ UNMATCH

          DISPLAY CORRECT
          JMP EXIT
UNMATCH: DISPLAY WRONG

EXIT:  INT 3H
CODE ENDS
 END START
```

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Conclusion: The program successfully demonstrates password verification in 8086 assembly language using DOS interrupt services and string handling techniques. It highlights the use of buffered input, custom string comparison, and structured control flow through subroutines. By utilizing string, data transfer, and control transfer instructions effectively, the program validates user input and displays appropriate access messages
of authentication logic.

Exercise 1:

Aim: To analyze the use of DOS interrupts for handling unbuffered keyboard input and screen output in 8086 assembly language.

Abstract: Develop an 8086 assembly language program that securely reads a 6-character password from the user and verifies it against a predefined password. As each character is entered, it should be immediately masked on the screen by displaying an 'X' to prevent revealing the actual input. The program must store the entered characters in a buffer and perform a character-by-character

comparison with the correct password using a custom subroutine. Based on the result of the comparison, the program should display either "Access Granted" or "Access Denied" and terminate gracefully.